

CS 102 Lab 6 – Extensions

Extension 1: (IDs ending with 3, 6, 9)

Write a recursive method that returns all unique big integers in an array. To do this, use an algorithm that is similar to merge sort algorithm, but change the merge step so that, duplicate values are removed. Your final method should return a new array containing only the unique big integers.

Extension 2: (IDs ending with 1, 4, 7)

Implement a variant of merge sort in which the array is divided into three parts instead of two. Compare the runtime of three-way merge sort algorithm with the two-way merge sort algorithm.

Extension 3: (IDs ending with 0, 2, 5, 8)

Print a histogram for the random big integers. The histogram should show how many big integers fall into each category based on their most significant digit.

- Given a large number of randomly generated numbers and a good random number generator, what should this distribution look like?
- What was the distribution of the output produced by John von Neumann's middle-square pseudorandom number generator?